PROJECT

CS-844: GENERATIVE DEEP MODELS

PAPER REVIEW: DENOISING DIFFUSION PROBABILISTIC MODELS

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1. Comprehensive Summary

Denoising Diffusion Probabilistic Models (DDPM), introduced by Ho, Jain, and Abbeel at NeurIPS 2020, is a landmark paper that established a practical and theoretically grounded framework for high-quality generative image synthesis using diffusion probabilistic models. The paper builds on earlier theoretical work by Sohl-Dickstein et al. (2015) and connects it to denoising score matching (Vincent, 2011) and Langevin dynamics, producing a method that achieves image quality competitive with, and in some cases surpassing, state-of-the-art Generative Adversarial Networks (GANs) at the time.

The central idea is to define a generative model through two coupled Markov chains: a fixed forward process and a learned reverse process. The forward (noising) process q gradually corrupts a data sample x0 by adding small amounts of Gaussian noise over T discrete time steps (T = 1000 in the paper's experiments). Specifically, at each step t, the noisy sample xt is produced by:

$$q(x_t \mid x_{t-1}) = N(x_t; \sqrt{(1-\beta_t)}\cdot x_{t-1}, \beta_t \cdot I)$$

where β_t is a variance schedule. A key analytical convenience is that xt can be sampled in closed form directly from x0 at any timestep using the reparameterization trick, without iterating through all intermediate steps. This allows efficient training.

The reverse process pθ is a learned neural network that attempts to undo the forward noising process step by step. Rather than directly predicting the denoised image, the authors make the critical simplification of training the network to predict the noise ε that was added at each step. This noise prediction objective a simple mean-squared error between the true noise and the network's prediction emerges naturally from a reweighted variational lower bound on the data likelihood.

The neural network backbone used for noise prediction is a U-Net architecture with residual blocks, group normalization, and self-attention layers at multiple spatial resolutions. Time-step information is injected into the network using sinusoidal positional embeddings (borrowing from Transformer architectures). The resulting model achieves an Inception Score (IS) of 9.46 and Frechet Inception Distance (FID) of 3.17 on CIFAR-10 unconditional generation, representing state-of-the-art quality at the time. On 256x256 LSUN and CelebA-HQ datasets, the generated samples are visually indistinguishable from real images in many cases.

Beyond image generation, the paper also demonstrates a connection between DDPM and lossy compression: the forward process can be interpreted as a progressive encoder, with the reverse process acting as the decoder. This connection provides a principled probabilistic interpretation that goes beyond the purely empirical motivation of many GAN-based methods.

2. Type of Generative Deep Model

DDPM belongs to the class of Latent Variable Generative Models, and more specifically to the family of Diffusion Probabilistic Models (also called score-based generative models in the related literature). Unlike GANs, which define an implicit distribution through adversarial training, or VAEs, which optimize a tractable lower bound on the likelihood using a recognition network, diffusion models define a parameterized Markov chain as the generative process.

Diffusion models are closely related to score-based generative models: the noise prediction network in DDPM is equivalent to learning the score (gradient of the log data density) at different noise levels, a connection formally established by Song and Ermon (2020) and later unified in the stochastic differential equations (SDE) framework of Song et al. (2021). DDPM can therefore also be described as a hierarchical VAE with a fixed (non-learnable) encoder, where each level of the hierarchy corresponds to one denoising step. The reverse process defines an extremely deep generative model 1000 steps deep each step involving a single forward pass through a U-Net.

3. Advantages Over Prior and Baseline Methods

Training stability. Unlike GANs, which require a delicate adversarial balance between the generator and discriminator, DDPM training is simple and stable. The loss is a straightforward mean-squared error and does not suffer from mode collapse, vanishing gradients, or the oscillatory dynamics that plague GAN training. This makes DDPM significantly easier to scale and reproduce.

Sample diversity. GANs are well known for mode dropping the generator learns to cover only a subset of the data distribution. DDPM, by contrast, defines an explicit probabilistic model and naturally produces diverse samples across all modes of the data distribution. This is confirmed by the superior Recall metrics (which measure coverage) reported in the paper.

Improved image quality over prior likelihood-based models. Previous state-of-the-art likelihood-based models such as standard VAEs, Flow-based models (Glow, RealNVP), and autoregressive models (PixelCNN++) often produced blurry or low-resolution images. DDPM generates sharp, high-fidelity images while also modeling an explicit likelihood (albeit through its ELBO), which GANs do not.

Connection to score matching. DDPM unifies two previously separate lines of work diffusion probabilistic models and denoising score matching providing a cleaner theoretical framework for understanding and extending generative models. This theoretical clarity enabled a wave of subsequent improvements.

Flexible conditioning. The architecture naturally supports conditional generation by conditioning the U-Net on class labels or other guidance signals, without requiring architectural changes like the conditional variants of GANs (cGAN, BigGAN).

4. Downsides and Limitations

- Slow sampling speed is the most significant limitation of DDPM. Generating a single image requires T = 1000 sequential forward passes through the U-Net. This makes inference approximately 1000x slower than a single-pass GAN at the same resolution, making real-time or interactive generation impractical on standard hardware.
- High computational cost during both training and inference. Training DDPM for high-resolution image generation requires substantial GPU memory and time. The U-Net must process full-resolution feature maps at multiple scales, and T = 1000 gradient steps must be accumulated through backpropagation.
- The model does not admit an encoder in the traditional sense. Unlike VAEs, which learn an explicit posterior over latent codes, DDPM's latent space is a fixed Gaussian noise vector reached through the forward process. This makes image editing, interpolation, and latent-space manipulation less straightforward.
- Likelihood evaluation is expensive. While DDPM defines an ELBO, evaluating the exact likelihood requires summing over all T diffusion steps, which is computationally prohibitive for downstream applications such as out-of-distribution detection.
- Discrete-time formulation. The original DDPM operates with a fixed number of discrete steps and a fixed variance schedule. This rigidity means the user cannot flexibly trade computation for sample quality without retraining the model.

5. Subsequent Improvements and Future Directions

The publication of DDPM triggered an enormous body of follow-up work, addressing most of the limitations described above.

Faster sampling. Denoising Diffusion Implicit Models (DDIM, Song et al., ICLR 2021) reformulate the reverse process as a non-Markovian chain, allowing high-quality generation in as few as 10–50 steps by reusing the same trained model. This yielded a 10–50x speed improvement with minimal quality degradation. Subsequent samplers such as PNDM, DPM-Solver, and DEIS further reduced the required steps to fewer than 20 in many cases.

Improved architecture and training. Nichol and Dhariwal (Improved DDPM, ICML 2021) introduced learned variance schedules, cosine noise scheduling, and importance sampling for the diffusion timesteps, significantly improving the ELBO and log-likelihood on standard benchmarks while also improving sample quality.

Classifier guidance and classifier-free guidance. Dhariwal and Nichol (Diffusion Models Beat GANs on Image Synthesis, NeurIPS 2021) demonstrated that diffusion models surpass GANs on ImageNet by using an ADM-U-Net architecture and classifier guidance. Ho and Salimans (2021) later introduced classifier-free guidance, which eliminates the need for a separate classifier by jointly training conditional and unconditional models, improving both sample quality and diversity.

Latent diffusion models. Rombach et al. (Latent Diffusion Models / Stable Diffusion, CVPR 2022) moved the diffusion process into the compressed latent space of a pre-trained autoencoder, drastically reducing computational cost while maintaining high sample quality. This is the foundation of Stable Diffusion and similar large-scale text-to-image systems.

Continuous-time SDE formulation. Song et al. (Score-Based Generative Modeling through SDEs, ICLR 2021 Outstanding Paper) unified DDPM and SMLD (score matching with Langevin dynamics) under a continuous-time SDE framework, enabling flexible noise schedules, exact likelihood computation, and controllable trade-offs between speed and quality.

Consistency models. Song et al. (Consistency Models, ICML 2023) introduced a new class of generative models that directly map noisy observations to clean samples in a single step by enforcing self-consistency along diffusion trajectories, achieving competitive quality with far fewer function evaluations.